

AWS Services

Service	Purpose
Amazon S3	Hosts the compiled Ionic/Angular frontend as a static website. The same deployment also uses S3 object storage for application image files.
AWS Elastic Beanstalk	Hosts the Node.js/Express backend API. The API is deployed from the compiled backend bundle.
Amazon RDS	Hosts the PostgreSQL database used by Sequelize models for users and feed items.
IAM	Provides deployment credentials used by CircleCI to publish builds to S3 and Elastic Beanstalk.

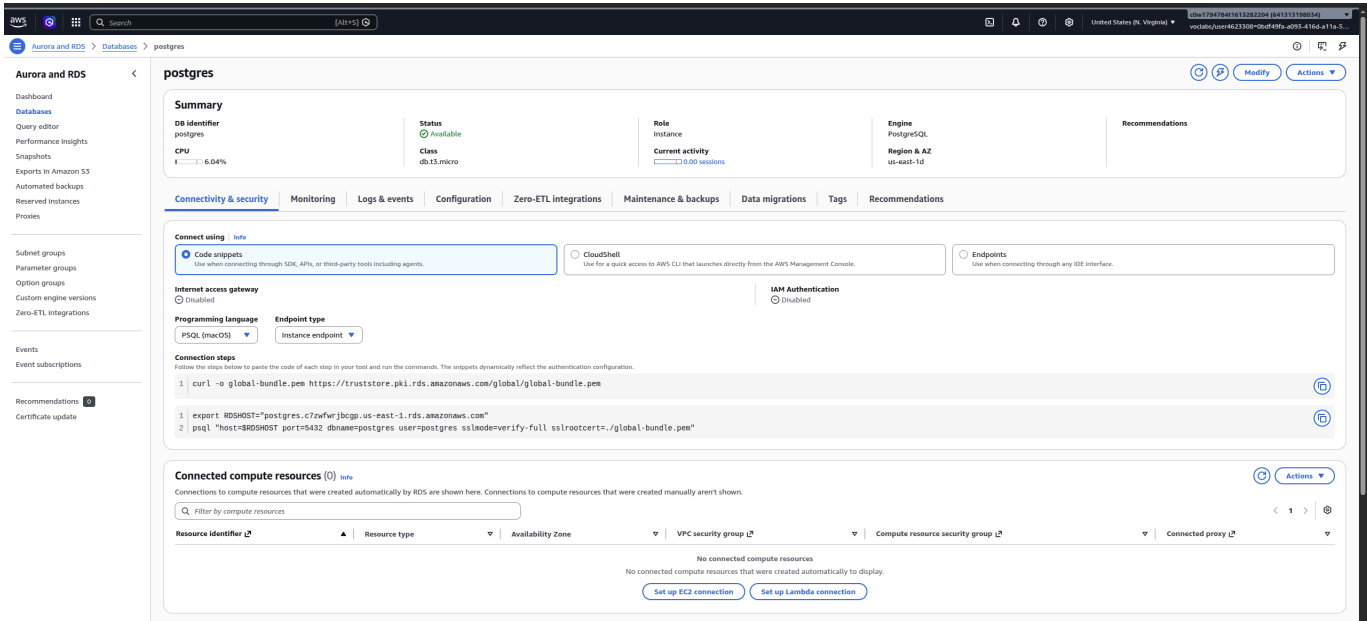
- Frontend S3 bucket: `mj-udacity-udagram-s3`
- Frontend URL: `http://mj-udacity-udagram-s3.s3-website-us-east-1.amazonaws.com`
- API Elastic Beanstalk environment: `Mj-udacity-udagram-api-env`
- API URL: `http://mj-udacity-udagram-api-env.eba-wcfdcn9n.us-east-1.elasticbeanstalk.com/api/v0`
- Database: PostgreSQL on Amazon RDS

See [Architecture.md](#).

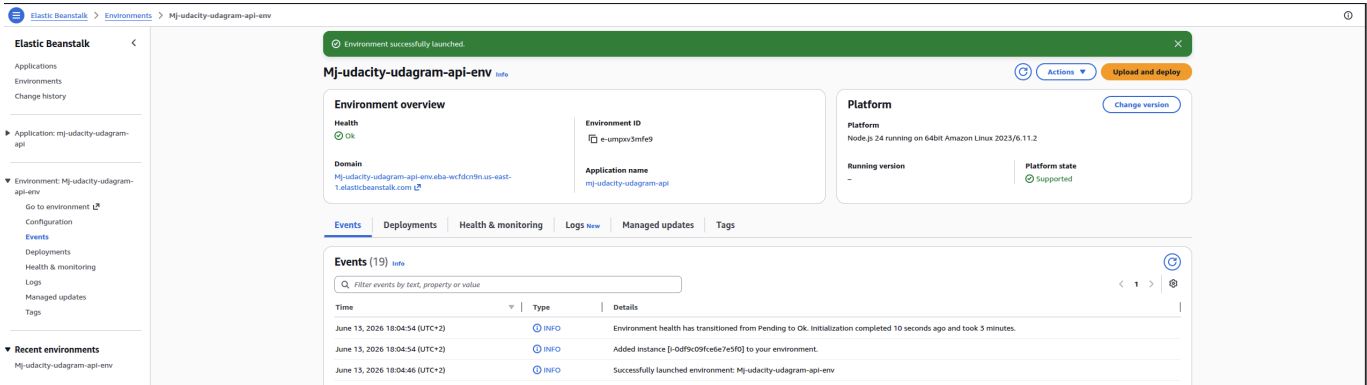
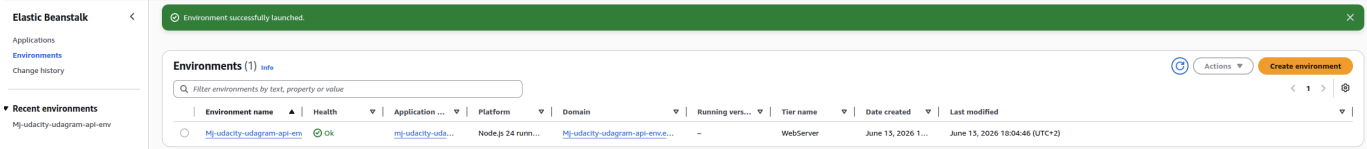
Infrastructure Screenshots

The screenshot shows the AWS Aurora and RDS console. The 'Databases' page is active, displaying a table of database instances. The table has columns for DB Identifier, Status, Role, Engine, Upgrade rollout order, Region, Size, Recommendations, CPU, Current activity, Maintenance, VPC, and Multi-AZ. A single instance named 'postgres' is listed with a status of 'Available'.

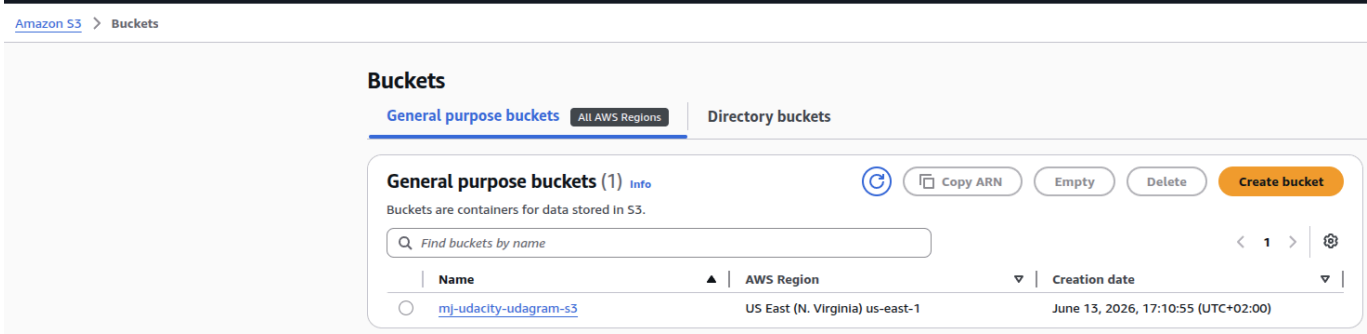
DB Identifier	Status	Role	Engine	Upgrade rollout order	Region	Size	Recommendations	CPU	Current activity	Maintenance	VPC	Multi-AZ
postgres	Available	Instance	PostgreSQL	SECOND	us-east-1d	db.t3.micro		5.35%	1.00 sessions	none	vpc-0d46e65f95766af	No



Elastic Beanstalk



S3



mj-udacity-udagram-s3

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Bucket overview

AWS Region

US East (N. Virginia) us-east-1

Amazon Resource Name (ARN)

arn:aws:s3::mj-udacity-udagram-s3

Creation date

June 13, 2026, 17:10:55 (UTC+02:00)

Bucket Versioning

Edit

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

Disabled

Multi-factor authentication (MFA) delete

An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)

Disabled

Bucket ABAC

Edit

Attribute-based access control (ABAC) is an authorization strategy that defines permissions based on attributes. With ABAC, you can attach tags to your general purpose buckets and AWS Identity and Access Management (IAM) entities (users or roles), then scale access to objects in your S3 general purpose buckets using tag-based policies. [Learn more](#)

ABAC status

Disabled

Default encryption

Edit

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type

Info

Server-side encryption with Amazon S3 managed keys (SSE-S3)

Bucket Key

When KMS encryption is used to encrypt new objects in this bucket, the bucket key reduces encryption costs by lowering calls to AWS KMS. [Learn more](#)

Enabled

Blocked encryption types

Info

Server-Side Encryption with Customer-Provided Keys (SSE-C)

Intelligent-Tiering Archive configurations (0)

View details

Edit

Delete

Create configuration

Enable objects stored in the Intelligent-Tiering storage class to tier-down to the Archive Access tier or the Deep Archive Access tier which are optimized for objects that will be rarely accessed for long periods of time. [Learn more](#)

Find Intelligent-Tiering Archive configurations

Name	Status	Scope	Days until transition to Archive Access t...	Days until transition to Deep Archive A...
No archive configurations No configurations to display.				

Create configuration

Server access logging

Edit

Log requests for access to your bucket. Use [CloudWatch](#) to check the health of your server access logging. [Learn more](#)

Server access logging

Disabled

AWS CloudTrail data events

You can view and configure CloudTrail data events for Amazon S3 bucket object-level operations in the AWS CloudTrail console.

AWS CloudTrail

Permissions overview

Access finding
Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#) 
[View analyzer for us-east-1](#)

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#) 

Block all public access
 Off

► Individual Block Public Access settings for this bucket

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#) 

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3::mj-udacity-udagram-s3/*"
    }
  ]
}
```

Environment Configuration

Environment-specific and secret values are not committed to source code. The production API loads configuration from environment variables through `udagram/udagram-api/src/config/config.ts`. CircleCI stores the required secrets as project environment variables for deployment, and the deploy job runs `eb setenv` to copy the backend runtime values into the Elastic Beanstalk environment properties. The running API process reads those values from Elastic Beanstalk at runtime, not from checked-in files.

Production values include:

- `POSTGRES_HOST`
- `POSTGRES_DB`
- `POSTGRES_USERNAME`
- `POSTGRES_PASSWORD`
- `AWS_BUCKET`
- `AWS_ACCESS_KEY_ID`
- `AWS_SECRET_ACCESS_KEY`
- `AWS_SESSION_TOKEN`
- `AWS_REGION`
- `JWT_SECRET`
- `URL`

Elastic Beanstalk production environment properties should include after CircleCI copy:

- `POSTGRES_HOST`
- `POSTGRES_DB`
- `POSTGRES_USERNAME`
- `POSTGRES_PASSWORD`

- `AWS_BUCKET`
- `AWS_ACCESS_KEY_ID`
- `AWS_SECRET_ACCESS_KEY`
- `AWS_SESSION_TOKEN`
- `AWS_REGION`
- `JWT_SECRET`
- `URL`

`AWS_ACCESS_KEY_ID`, `AWS_SECRET_ACCESS_KEY`, and `AWS_SESSION_TOKEN` must belong to the same temporary AWS credentials session. If the session expires or only one of the three values is updated, the backend can still generate signed S3 URLs, but S3 rejects uploads with `InvalidToken`.